# ECE276A Project 1 Report: Building a Panorama From IMU Data

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***Abstract*—This project report focuses on the methods used to reconstruct a camera panoramic from Inertial Measurement Unit (IMU) data. In robotics, cameras deal with varying conditions that decrease reliability of feature-based stitching algorithms. Therefore, we introduce a method where we estimate camera orientation directly from inertial measurements. We use projected gradient descent with a motion model, restricting to the unit quaternions, to optimize the trajectories. The final quaternions can be used to build rotation matrices that map camera images onto a unit sphere, which can be projected onto a cylinder and unrolled with equidistant projection. This gives our final panorama constructed from pure IMU data.**

## I. Introduction

Panoramic imaging produces a wide field-of-view representation of a scene by stitching together multiple camera frames taken over time. In many robotics settings, however, reliable image-based stitching is difficult as the scene may lack distinct visual features, lighting can change, and motion blur can decrease correspondence, which impacts the stitching algorithm. This can impair many fundamental components of robotics perception such as visual odometry, SLAM, and navigation.

To address these limitations, we use gyroscope and accelerometer measurement from an Inertial Measurement Unit (IMU) to approximate the camera's quaternion rotation over time. The estimated orientation is then used to align the images and stitch them into a panorama. The end goal is to reduce reliance on feature matching and improve reliability and stability of the resulting panorama.

Our approach estimates orientation trajectories by representing rotations as unit quaternions and optimizing them with projected gradient descent. We define a cost objective with:

- a motion model formed by integrating gyroscope measurements.
- an observation model formed by comparing world frame acceleration due to gravity with IMU acceleration measurements.

Each gradient step is followed by projecting back to the unit quaternions to maintain valid rotations. After estimating the quaternion orientation, we convert each quaternion to a rotation matrix that maps from body frame to world frame. Each image pixel is first mapped to a unit-sphere using the camera field of view, then rotated into the world frame, and finally projected to a cylinder and unrolled with equirectangular projection. This produces a panorama based solely on optimized IMU data.

## II. Problem Formulation

Let $\{I_k\}_{k=1}^N$ be the sequence of camera images captured with timestamps $\{t_k\}_{k=1}^N$. The IMU provides timestamped gyroscope and accelerometer measurements

$$\mathbf{x}_t = \begin{bmatrix}\tau_t & \mathbf{a}_t^T & \omega_t^T\end{bmatrix}^T, \ \mathbf{a}_t \in \mathbb{R}^3, \ \omega_t \in \mathbb{R}^3 \tag{1}$$

where $\tau_t$ is the IMU timestamp, $\mathbf{a}_t$ is the measured acceleration, and $\omega_t$ is the measured angular velocity.

### A. Timestamp Association

Because camera and IMU are sampled asynchronously, each image must be paired with an orientation. We define a nearest-neighbor timestamp association function

$$g(t_k) := \max\{t \in \{1, \dots, T\} \ : \ \tau_t \leq t_k\} \tag{2}$$

and assign image $I_k$ the quaternion $\mathbf{q}_{g(t_k)}$. The goal is to compute an optimized quaternion orientation $\{\mathbf{q}_t\}_{t=1}^T$ such that the associated orientations $\mathbf{q}_{g(t_k)}$ are accurate for projecting each image into the world frame.

### B. Motion Model

Let $\circ$ denote quaternion multiplication, $(\cdot)^{-1}$ for quaternion inverse, and $\Delta\tau_t = \tau_{t+1} - \tau_t$ be the time difference between two timestamps. Using the IMU angular velocity measurements $\omega_t$, we can use quaternion kinematics motion model to predict the quaternion at the next time step:

$$\mathbf{q}_{t+1} = f(\mathbf{q}_t, \Delta\tau_t\omega_t) := \mathbf{q}_t \circ \exp(\begin{bmatrix}0, \Delta\tau_t\omega_t/2\end{bmatrix}) \tag{3}$$

where $\exp(\cdot)$ is the quaternion exponential mapping of an axis-angle into a unit quaternion.

### C. Observation Model

Since the body undergoes pure rotation, the force measured by the accelerometer is dominated by gravity. In gravity units, the gravity direction in the world frame is $\begin{bmatrix}0, 0, 0, 1\end{bmatrix}$. If we rotate this to the body frame, we get $\begin{bmatrix}0, \mathbf{a}_t\end{bmatrix}$. Therefore, our observation model becomes:

$$[0, \mathbf{a}_t] = h(\mathbf{q}_t) := \mathbf{q}_t^{-1} \circ [0, 0, 0, 1] \circ \mathbf{q}_t \tag{4}$$

### D. Constrained Optimization

We estimate the full quaternion by minimizing the cost function

$$c(\mathbf{q}_{1:T}) = \frac{1}{2}\sum_{t=0}^{T-1} \left\| 2\,\log\big(\mathbf{q}_{t+1}^{-1} \circ f(\mathbf{q}_t, \Delta\tau_t\omega_t)\big)\right\|_2^2 + \frac{1}{2}\sum_{t=1}^{T} \| \,[0, \mathbf{a}_t] - h(\mathbf{q}_t)\|_2^2 \tag{5}$$

subject to the unit-norm constraint:

$$\min_{\mathbf{q}_{1:T}} c(\mathbf{q}_{1:T}) \quad \text{s.t.} \quad \|\mathbf{q}_t\|_2 = 1,\ \forall t \tag{6}$$

where $c(\mathbf{q}_{1:T})$ (5) penalizes deviations from the IMU motion and observation model. We solve (6) using projected gradient descent, where after each step, we project each $\mathbf{q}_t$ back onto the unit quaternions:

$$\mathbf{q}_t \leftarrow \frac{\mathbf{q}_t}{\|\mathbf{q}_t\|_2} \tag{7}$$

### E. *Problem*

Given IMU measurements $\{(\tau_t, \mathbf{a}_t, \omega_t)\}_{t=1}^{T}$ and camera images $\{(t_k, I_k)\}_{k=1}^{N}$, estimate a unit-quaternion orientation $\mathbf{q}_{1:T}$ directly from $(\mathbf{a}_t, \omega_t)$ by solving a constrained optimization problem

$$\min_{\mathbf{q}_{1:T}} c(\mathbf{q}_{1:T}) \quad \text{s.t.} \quad \|\mathbf{q}_t\|_2 = 1 \quad \forall t \tag{8}$$

optimized under the cost function:

$$c(\mathbf{q}_{1:T}) = \frac{1}{2}\sum_{t=0}^{T-1} \left\| 2\,\log\big(\mathbf{q}_{t+1}^{-1} \circ f(\mathbf{q}_t, \Delta\tau_t\omega_t)\big)\right\|_2^2 + \frac{1}{2}\sum_{t=1}^{T} \| \,[0, \mathbf{a}_t] - h(\mathbf{q}_t)\|_2^2 \tag{9}$$

For each image $I_k$, assign the closest-timestamp orientation

$$g(t_k) := \max\{t \in \{1, \dots, T\} \ :\ \tau_t \le t_k\},$$

and use $\mathbf{q}_{g(t_k)}$ to project the image into a common world reference frame for panorama construction.

## III. Technical Approach

### A. IMU Calibration

We calibrate the IMU by assuming that the first $T$ seconds are static. Under this static assumption, the accelerometer should measure only gravity,

$$\mathbf{a}^* = \begin{bmatrix} 0 & 0 & g \end{bmatrix}^T.$$

Let the raw accelerometer measurement at time $t$ be

$$\mathbf{a}_t = \mathbf{a}^\star + \beta_a + \mathbf{n}_t,$$

where $\beta_a \in \mathbb{R}^3$ is a constant accelerometer bias and $\mathbf{n}_t$ is zero-mean sensor noise. Assuming independent Gaussian noise,

$$\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \Sigma^2),$$

each component of $\mathbf{a}_t$ is distributed as

$$a_{x,t} \sim \mathcal{N}(\beta_{a,x}, \sigma^2), \tag{10}$$

$$a_{y,t} \sim \mathcal{N}(\beta_{a,y}, \sigma^2), \tag{11}$$

$$a_{z,t} \sim \mathcal{N}(g + \beta_{a,z}, \sigma^2). \tag{12}$$

Let $S := \{t : \tau_t \le T\}$ denote the set of indices within the static window. The MLE of the bias in the Gaussian model is the sample mean offset from $\mathbf{a}^\star$:

$$\hat{\beta}_a = \frac{1}{|S|}\sum_{t\in S} \mathbf{a}_t - \mathbf{a}^\star.$$

Finally, we correct the accelerometer bias by subtracting the estimated bias:

$$\mathbf{a}_t = \mathbf{a}_t - \hat{\beta}_a.$$

### B. Orientation Tracker

We begin by estimating the orientation of the quaternions $\mathbf{q}_{1:T}$ by using gyroscopic integration. Then, we compute the cost of the new estimates and apply projected gradient descent on the cost.

*a) GyroIntegration:* Here, we provide the algorithm used to estimate the quaternions orientation at time $t$.

**Algorithm 1** Gyroscope integration for starting quaternions

```
function GyroIntegration({τ_t}_{t=1}^T, {ω_t}_{t=1}^T)
    q_1 ← [1, 0, 0, 0]^⊤
    for t ← 1 to T − 1 do
        Δτ_t ← τ_{t+1} − τ_t
        q_{t+1} ← q_t ∘ exp([0, (Δτ_t/2) ω_t])
        q_{t+1} ← q_{t+1} / ||q_{t+1}||_2
    end for
    return q_{1:T}
end function
```

In this algorithm $\tau_t$ refers to the timestamp at $t$ and $\omega_t \in \mathbb{R}^3$ refers to the angular velocities.

*b) ComputeCost:* Next, we have the compute cost function. This function computes the error for every quaternion and sums the contribution from each.

**Algorithm 2** Compute cost function

```
function ComputeCost(q_{1:T}, ω, a, Δτ)
    c_mot ← 0,  c_obs ← 0
    for t ← 1 to T − 1 do
        q_{t+1} ← q_t ∘ exp([0, (Δτ_t/2) ω_t])
        c_mot ← c_mot + ||2 log(q_{t+1}^{-1} ∘ q_{t+1})||_2^2
        â_{t+1} ← (q_{t+1}^{-1} ∘ [0, 0, 0, 1] ∘ q_{t+1})
        c_obs ← c_obs + ||a_{t+1} − â_{t+1}||
    end for
    return (1/2) c_mot + (1/2) c_obs
end function
```

Here, $\omega$ is angular velocity $\mathbf{a}$ is the IMU acceleration measurement, $\Delta\tau$ is the difference in timestamps.

*c) PGD Optimization:* Our last function is the optimization algorithm using projected gradient descent [2]. The algorithm initializes with the gyroscope estimates, then runs PGD by computing the cost, taking a step towards the gradient $\nabla c$ determined by the Jacobian. We then project to $\mathbb{H}_*$, the unit quaternions.

**Algorithm 3** Optimization algorithm using previous functions

**function** OPTIMIZE($\{\tau_t, \omega_t, \mathbf{a}_t\}_{t=1}^{T},\ \eta,\ I,\ \epsilon$)
  $\Delta\tau_t \leftarrow \tau_{t+1} - \tau_t,\ \forall t = 1, \ldots, T-1$
  $\mathbf{q}_{1:T}^{(0)} \leftarrow$ GYROINTEGRATION($\{\tau_t\}, \{\omega_t\}$)
  $\mathbf{q}_{1:T} \leftarrow \mathbf{q}_{1:T}^{(0)}$
  $c^{(0)} \leftarrow +\infty$
  **for** $i \leftarrow 1$ to $I$ **do**
    $c^{(i)} \leftarrow$ COMPUTECOST($\mathbf{q}_{1:T}, \{\omega_t\}, \{\mathbf{a}_t\}, \{\Delta\tau_t\}$)
    **if** $|c^{(i)} - c^{(k-1)}| < \epsilon$ **then**
      **return** $\mathbf{q}_{1:T}$
    **end if**
    $\nabla c \leftarrow \frac{\partial}{\partial \mathbf{q}_{1:T}} c^{(i)}$ ▷ autograd Jacobian
    $\mathbf{q}_{1:T} \leftarrow \mathbf{q}_{1:T} - \eta\, \nabla c$
    **for** $t \leftarrow 1$ to $T$ **do**
      $\mathbf{q}_t \leftarrow \mathbf{q}_t / \|\mathbf{q}_t\|_2$ ▷ projection to $\mathbb{H}_*$
    **end for**
  **end for**
  **return** $\mathbf{q}_{1:T}$
**end function**

In this function, $\{\tau_t, \omega_t \mathbf{a}_t\}$ are IMU readings, $\eta$ is the step size, $I$ is the max iterations, and $\epsilon$ is the convergence threshold.

### C. Panorama

Let $y_k \in Y$ be any camera image with corresponding timestamp $t_k \in T_Y$. We begin by determining the offsets for each image.

*a) Determining the offset:* Let $y_k$ have width $W$ and height $H$, and let the camera horizontal/vertical fields-of-view be $\text{FOV}_h$ and $\text{FOV}_v$ in radians. For each pixel coordinate $(u, v) \in y_k$ with $u \in \{0, \ldots, W-1\}$ and $v \in \{0, \ldots, H-1\}$, define the angular offsets from the optical axis:

$$\lambda(u) = \left(\frac{u}{W-1} - \frac{1}{2}\right) \text{FOV}_h, \phi(v) = \left(\frac{v}{H-1} - \frac{1}{2}\right) \text{FOV}_v \tag{13}$$

Then, $\lambda, \phi \in [-0.5, 0.5]$. We then map these spherical angles to a cartesian ray in the body frame:

$$\mathbf{r}(u, v) = \begin{bmatrix} \cos\phi(v) \cos\lambda(u) \\ \cos\phi(v) \sin\lambda(u) \\ \sin\phi(v) \end{bmatrix} \in \mathbb{R}^3. \tag{14}$$

Let $\mathbf{r}(u, v)$ be a row $i \in R_{\text{body}}^{(k)}$. Then, $R_{\text{body}}^{(k)} \in \mathbb{R}^{(HW)\times 3}$ is the collection of cartesian rays for one image.

*b) Timestamp alignment:* We now find the closest associated quaternion $\mathbf{q}_{g(t_k)}$ for image $y_k$ with associated timestamp $t_k$. Define $g(t_k)$:

$$g(t_k) := \max\{t \in \{1, \ldots, T\} \ :\ \tau_t \le t_k\} \tag{15}$$

Let $\mathbf{R}_{g(t_k)} \in SO(3)$ be the rotation matrix associated with $\mathbf{q}_{g(t_k)}$ [3]. We use this to rotate all camera rays into the world frame:

$$R_{\text{world}}^{(i)} = R_{\text{body}}^{(i)} \mathbf{R}_{g(t_k)}^T \in \mathbb{R}^{(HW)\times 3}. \tag{16}$$

Let $\mathbf{r}_w = [x, y, z]^T \in R_{\text{world}}$ denote a rotated ray in the world frame.

*c) Equirectangular mapping:* We convert each world ray $\mathbf{r}_w$ to longitude and latitude:

$$\ell = \text{atan2}(y, x), \qquad \beta = \arcsin(\text{clip}(z, -1, 1)), \tag{17}$$

where $\ell \in (-\pi, \pi]$ and $\beta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. For a panorama of width $W_p$ and height $H_p$, the corresponding pixel indices are

$$u_p = \left\lfloor \frac{\ell + \pi}{2\pi} W_p \right\rfloor, \qquad v_p = \left\lfloor \frac{\beta + \frac{\pi}{2}}{\pi} H_p \right\rfloor \tag{18}$$

This maps $u_p \in [0, W_p]$ and $v_p \in [0, H_p]$, implementing the equirectangular projection [1].

*d) Color assignment:* Let $y_k(u, v) \in \mathbb{R}^3$ be the RGB color at pixel $(u, v)$ in frame $k$. The panorama is constructed by assigning each projected pixel to its computed panorama bin:

$$P(v_p, u_p) = y_k(u, v) \tag{19}$$

where $P \in \mathbb{R}^{H_p \times W_p \times 3}$ is the panorama image.

## IV. RESULTS

### A. Comparison graphs for IMU set 1-9

These are the roll, pitch, yaw graphs for datasets 1-9, which include the unoptimized orientations

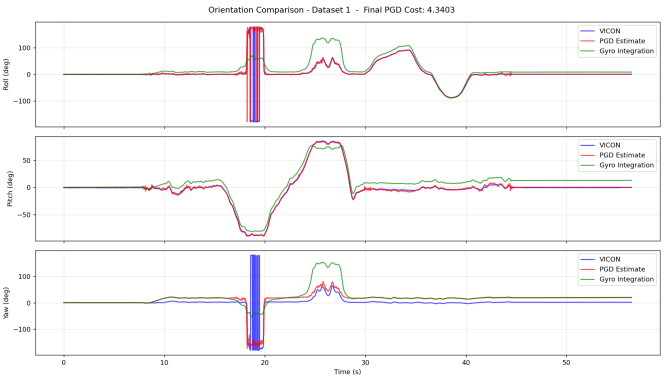



Fig. 1. Optimized orientation (1) compared with VICON.

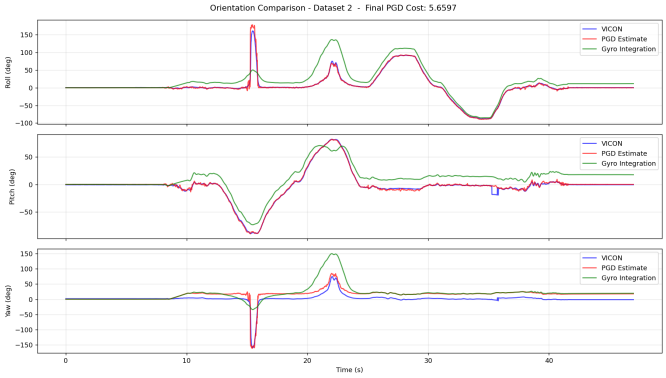



Fig. 2. Optimized orientation (2) compared with VICON.

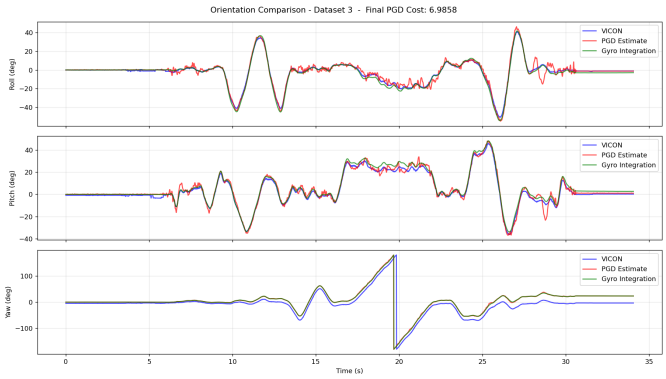



Fig. 3. Optimized orientation (3) compared with VICON.

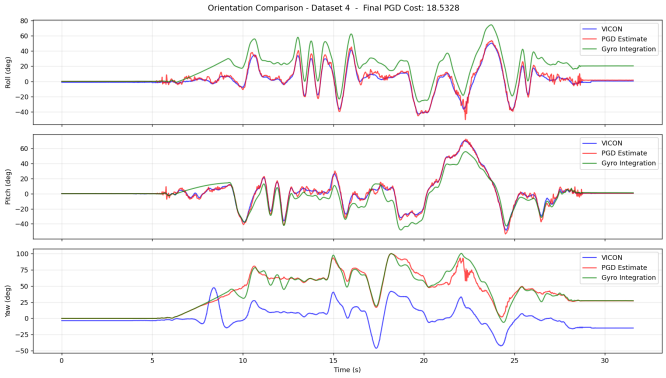



Fig. 4. Optimized orientation (4) compared with VICON.

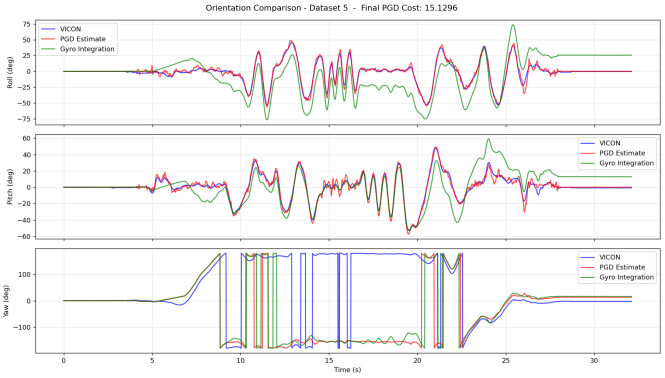



Fig. 5. Optimized orientation (5) compared with VICON.

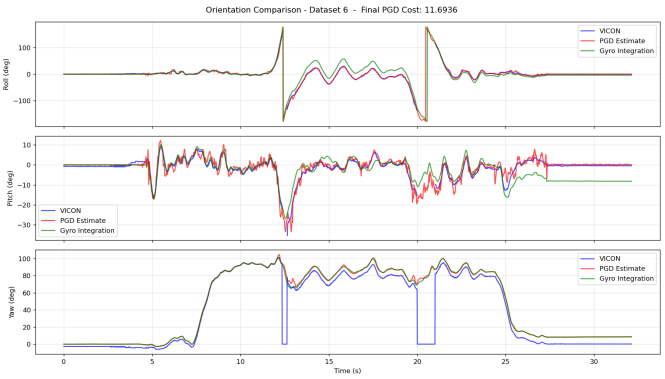



Fig. 6. Optimized orientation (6) compared with VICON.

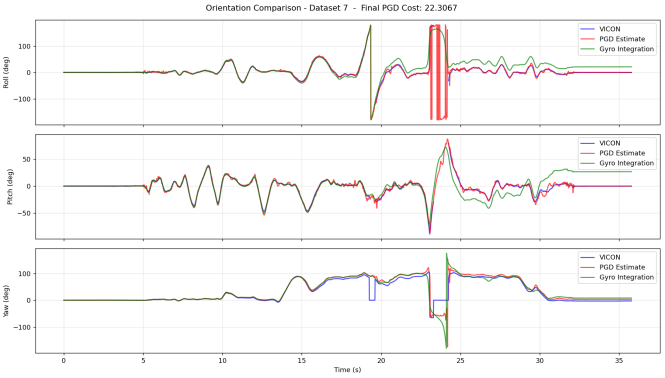



Fig. 7. Optimized orientation (7) compared with VICON.

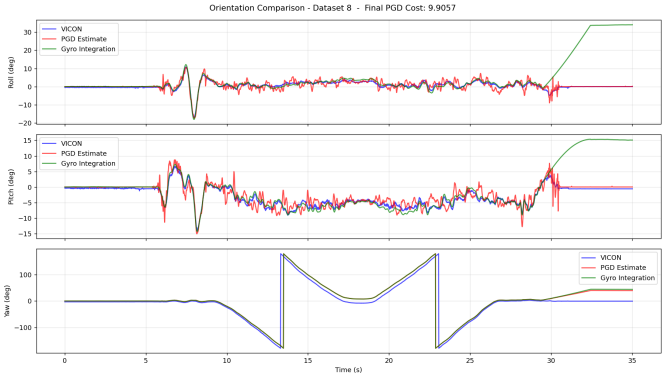



Fig. 8. Optimized orientation (8) compared with VICON.

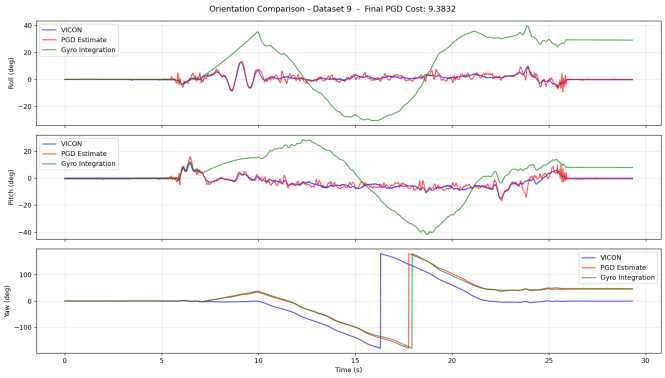



Fig. 9. Optimized orientation (9) compared with VICON.

## B. Comparison graphs for IMU set 10-11

These are the roll, pitch, yaw graphs for IMU datasets 10-11, with no VICON data.

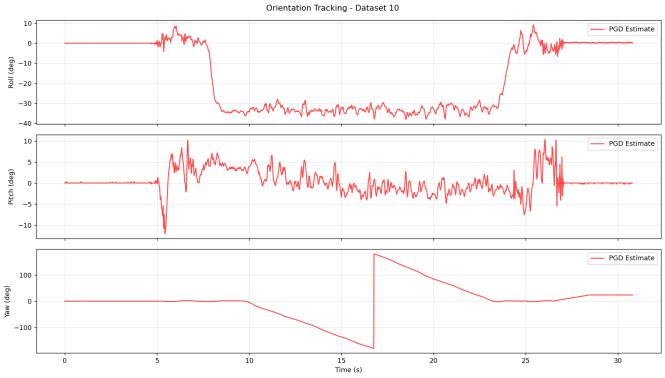



Fig. 10. Optimized orientation (10).

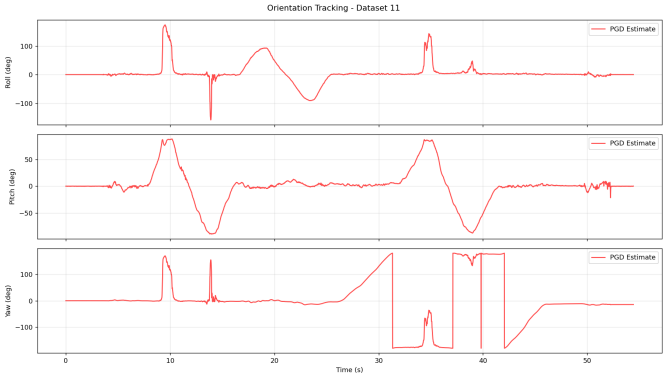



Fig. 11. Optimized orientation (11).

## C. Panorama Results for 1, 2, 8, 9

These are the panorama results from IMU datasets 1, 2, 8, 9. The VICON dataset is also included for comparison.

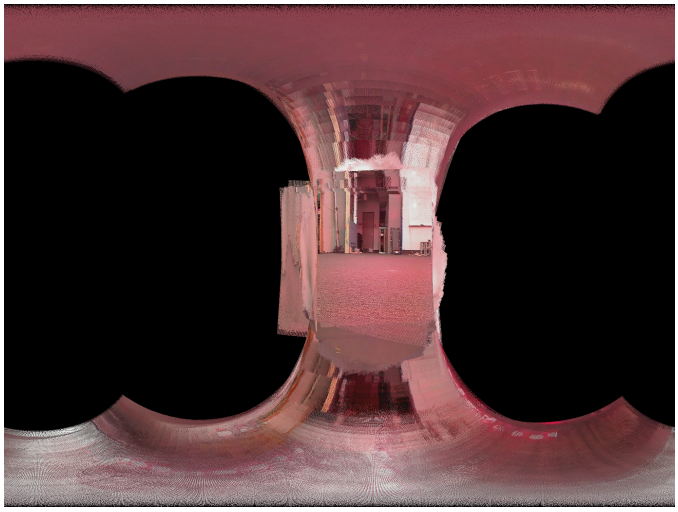


Fig. 12. Panorama without VICON (1).

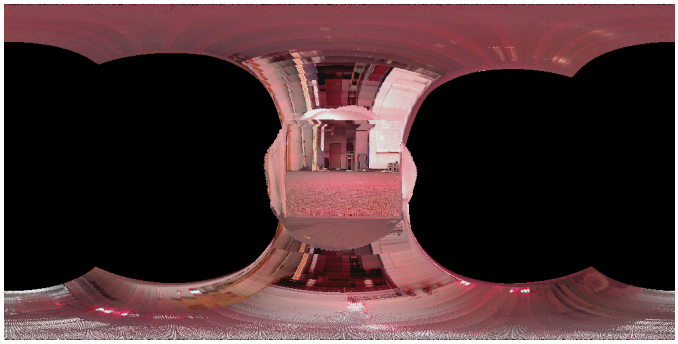


Fig. 13. Panorama with VICON (1).

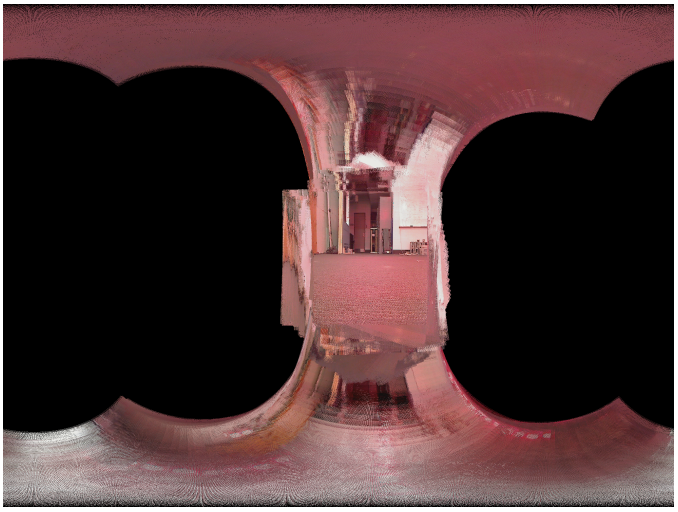


Fig. 14. Panorama without VICON (2).

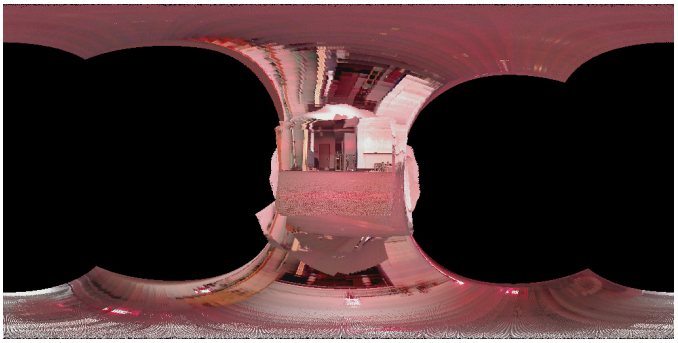


Fig. 15. Panorama with VICON (2).



Fig. 16. Panorama without VICON (8).



Fig. 17. Panorama with VICON (8).



Fig. 18. Panorama without VICON (9).



Fig. 19. Panorama with VICON (9).

### *D. Panorama Results for 10, 11*

These are the panorama results from IMU datasets 10, 11. There is no VICON comparison.

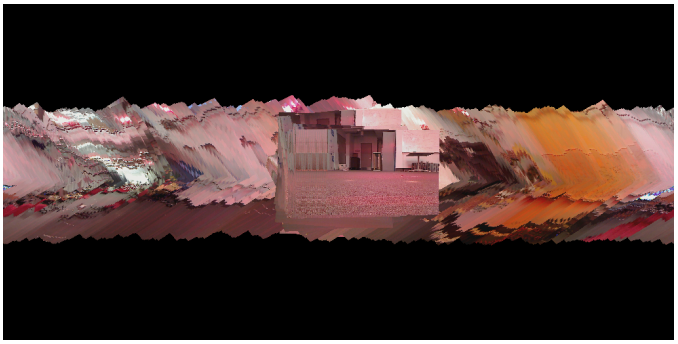


Fig. 20. Panorama without VICON (10).



Fig. 21. Panorama without VICON (11).

### *E. Discussion*

Looking at the resulting comparison graphs from figures (1) to (9), it is clear that PGD had significant impact on pitch and roll accuracy compared to the unoptimized gyroscope integration. This can also be seen by the similaritities between panoramas in (12) and (13).

Some limitations to the PGD orientation optimizer is the large error in the yaw angle as seen in figures (1) to (9). The observation model constrains orientation using the direction of gravity in the IMU frame. This constraint is informative for roll and pitch, since rotating about those axes changes the gravity direction. However, yaw corresponds to rotation about the gravity axis, which does not change the gravity direction. As a result, the observation term provides little to no information for yaw, and the optimizer must rely almost entirely on the motion model to determine that component.

We also see that the optimized quaternion trajectory can be noisy in certain areas rather than smoothly varying over time. Since full-batch projected gradient descent was used directly on $\{\mathbf{q}_t\}_{t=1}^{T}$, small measurement noise between the motion and observation likely produce variations in the estimated orientation. This impacted the quality of the final panorama by introducing distortions and ghosting.

Finally, the panorama construction itself may have also produced artifacts. We performed a direct equirectangular projection and color assignment without using a stitching algorithm. This caused cuts and distortions when similar images met.

## REFERENCES


[1] "Equirectangular projection," Wikipedia, The Free Encyclopedia. Accessed: Feb. 4, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Equirectangular_projection

[2] N. Atanasov, "ECE276A: Sensing & Estimation in Robotics Lecture 2: Unconstrained Optimization." Accessed: Feb. 2, 2026. [Online]. Available: https://natanaso.github.io/ece276a/ref/ECE276A_2_UnconstrainedOptimization.pdf

[3] N. Atanasov, "ECE276A: Sensing & Estimation in Robotics Lecture 3: Rotations." Accessed: Feb. 06, 2026. [Online]. Available: https://natanaso.github.io/ece276a/ref/ECE276A_3_Rotations.pdf